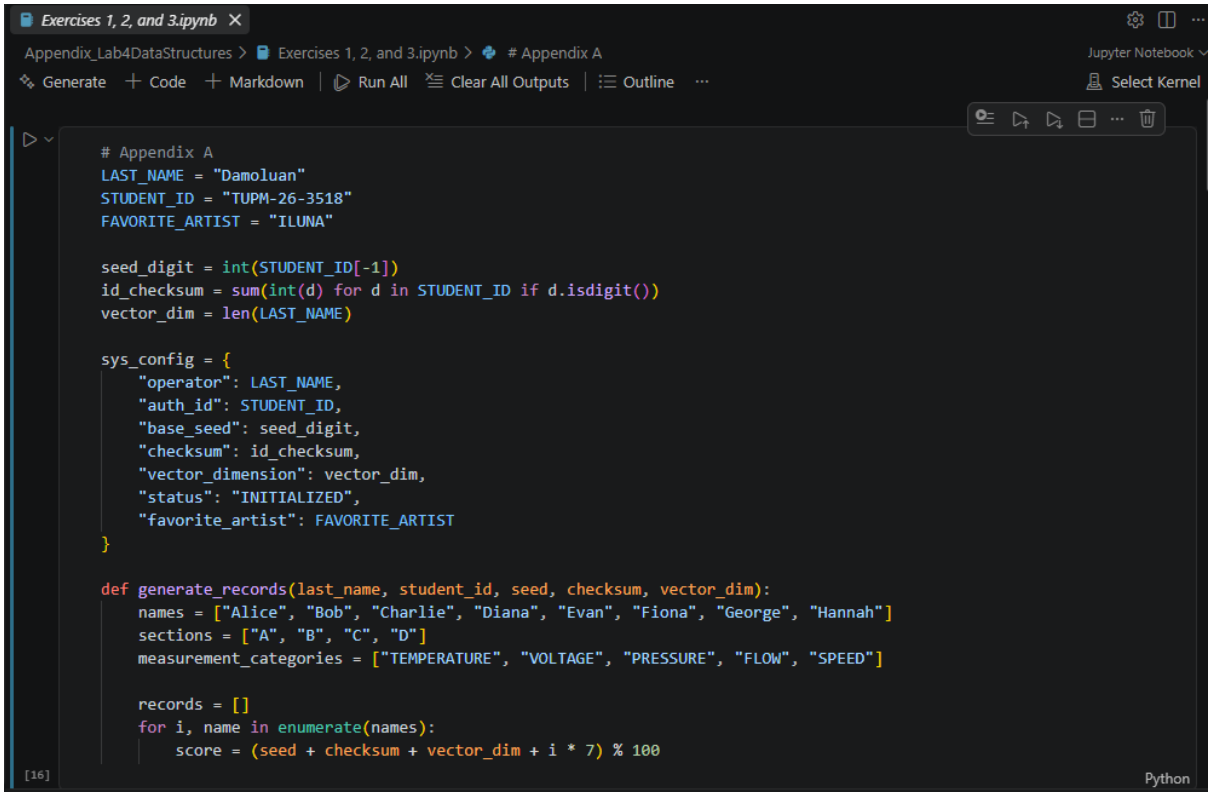
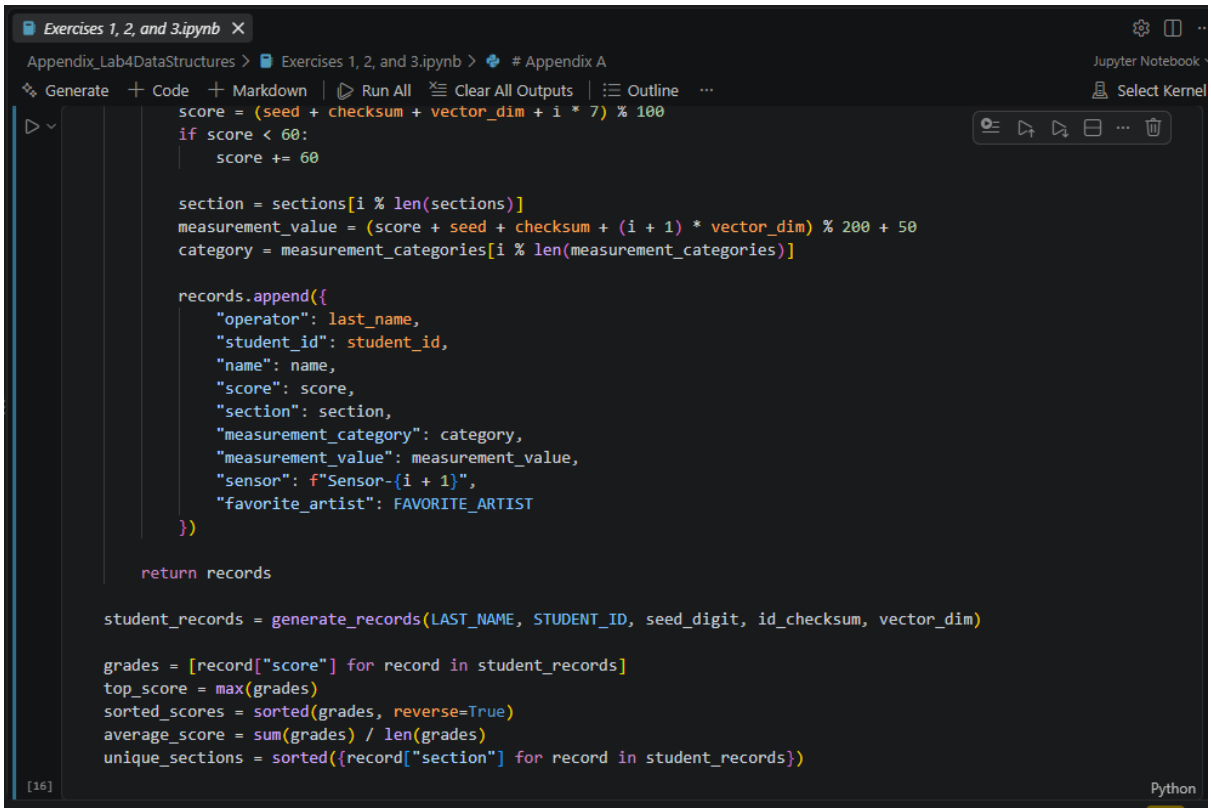
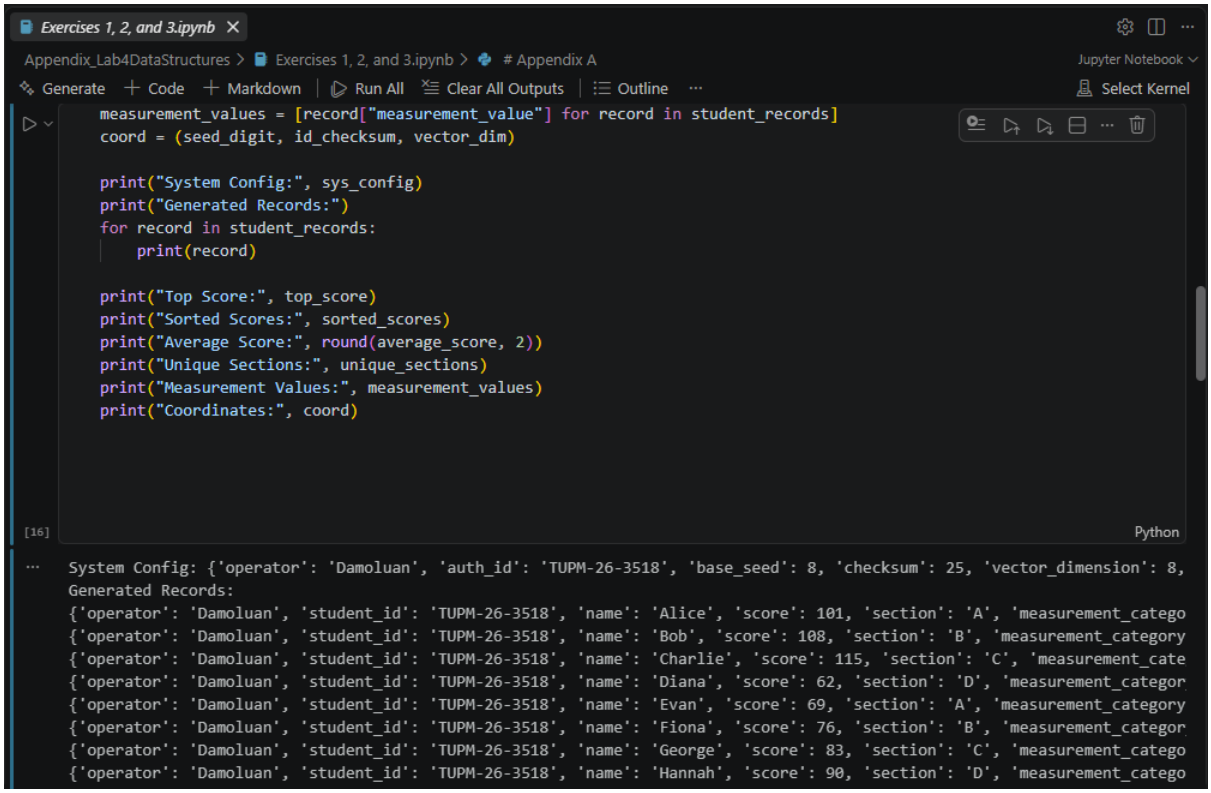
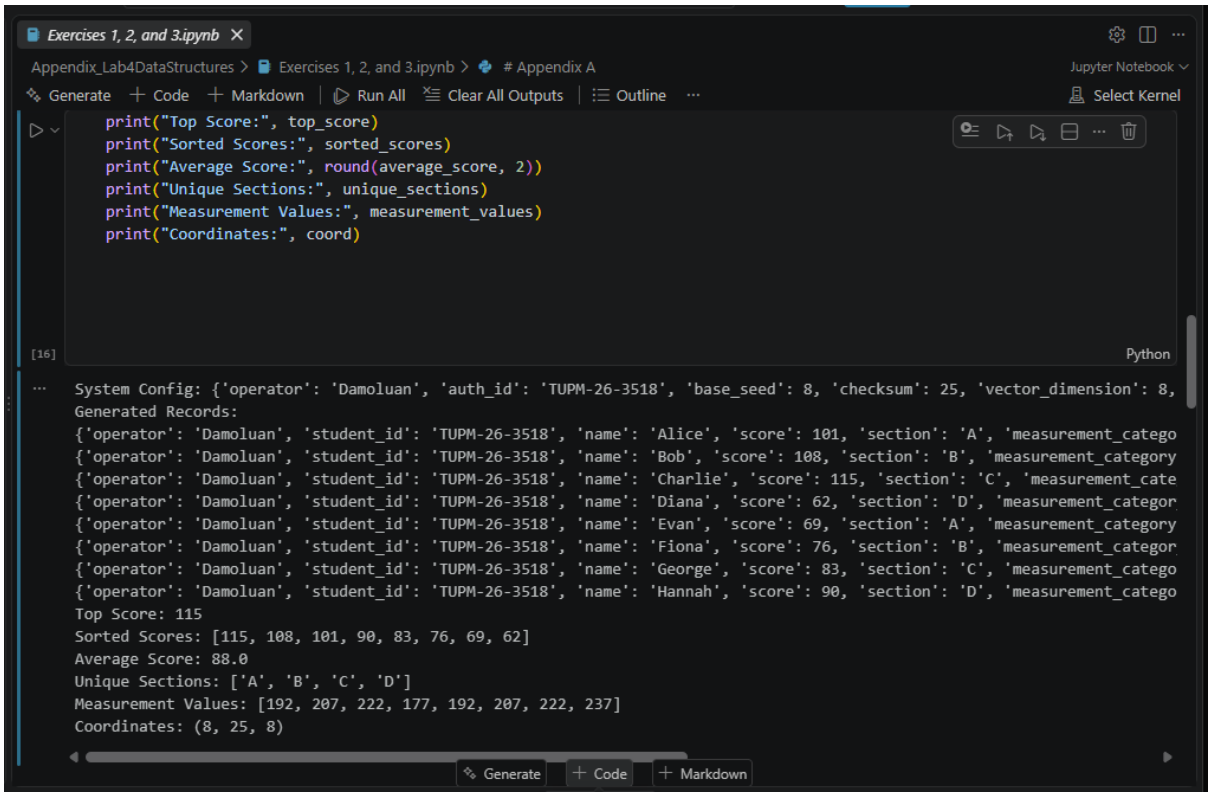
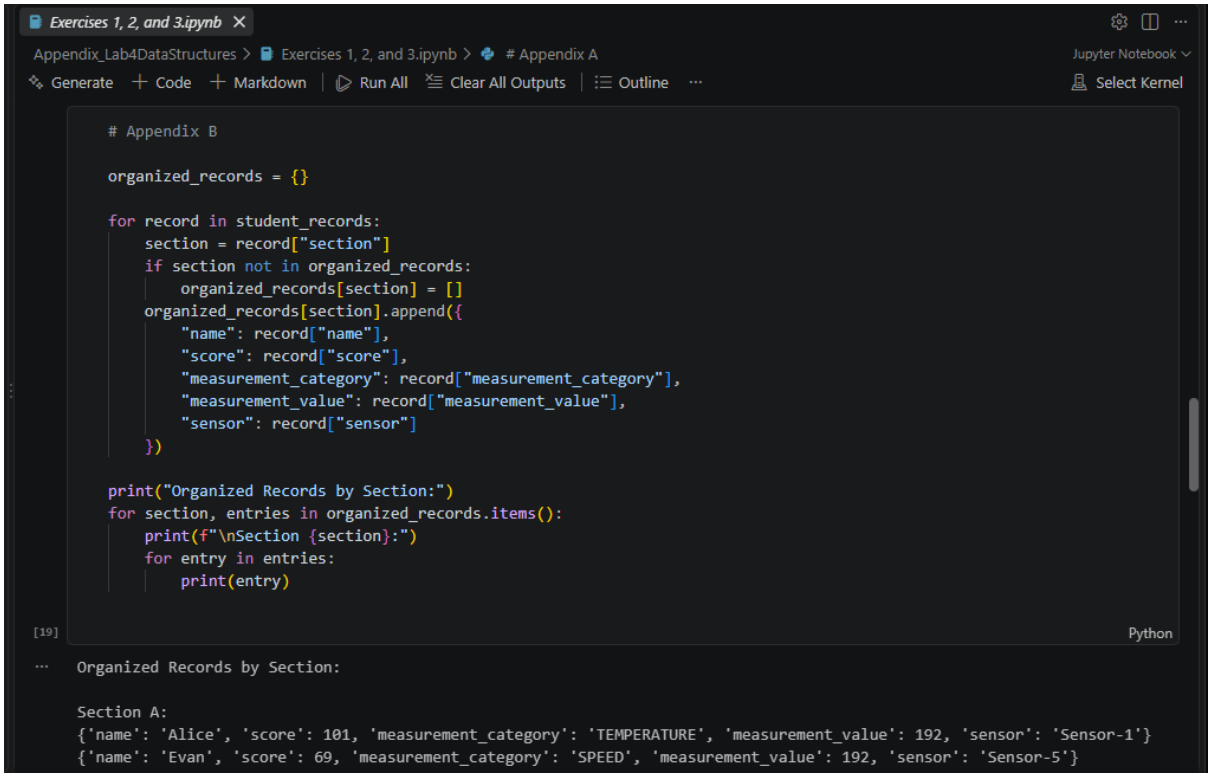
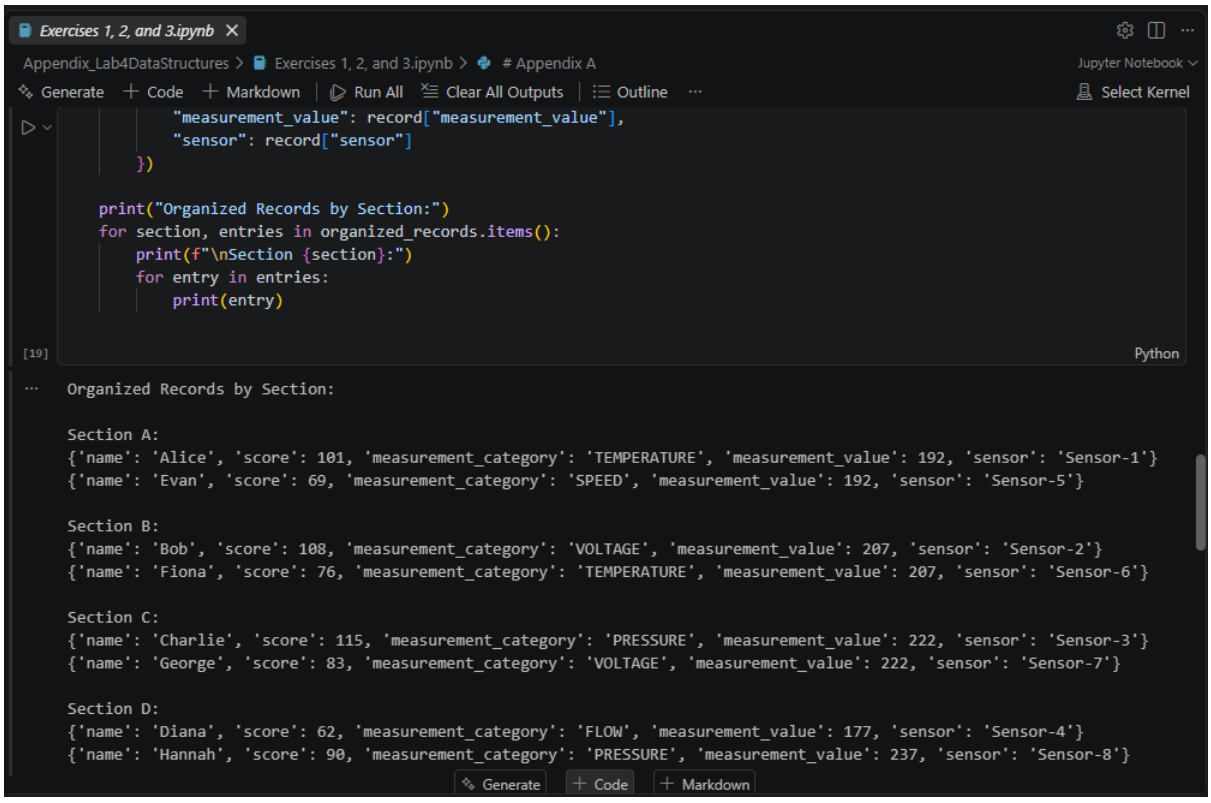

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Appendix A
LAST_NAME = "Damoluan"
STUDENT_ID = "TUPM-26-3518"
FAVORITE_ARTIST = "ILUNA"
seed_digit = int(STUDENT_ID[-1])
id_checksum = sum(int(d) for d in STUDENT_ID if d.isdigit())
vector_dim = len(LAST_NAME)
sys_config = {
"operator": LAST_NAME,
"auth_id": STUDENT_ID,
"base_seed": seed_digit,
"checksum": id_checksum,
"vector_dimension": vector_dim,
"status": "INITIALIZED",
"favorite_artist": FAVORITE_ARTIST
}
def generate_records(last_name, student_id, seed, checksum, vector_dim):
names = ["Alice", "Bob", "Charlie", "Diana", "Evan", "Fiona", "George", "Hannah"]
sections = ["A", "B", "C", "D"]
measurement_categories = ["TEMPERATURE", "VOLTAGE", "PRESSURE", "FLOW", "SPEED"]
records = []
for i, name in enumerate(names):
score = (seed + checksum + vector_dim + i * 7) % 100
[16]
Python


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score = (seed + checksum + vector_dim + i * 7) % 100
if score < 60:
score += 60
section = sections[i % len(sections)]
measurement_value = (score + seed + checksum + (i + 1) * vector_dim) % 200 + 50
category = measurement_categories[i % len(measurement_categories)]
records.append({
"operator": last_name,
"student_id": student_id,
"name": name,
"score": score,
"section": section,
"measurement_category": category,
"measurement_value": measurement_value,
"sensor": f"Sensor-{i + 1}",
"favorite_artist": FAVORITE_ARTIST
})
return records
student_records = generate_records(LAST_NAME, STUDENT_ID, seed_digit, id_checksum, vector_dim)
grades = [record["score"] for record in student_records]
top_score = max(grades)
sorted_scores = sorted(grades, reverse=True)
average_score = sum(grades) / len(grades)
unique_sections = sorted({record["section"] for record in student_records})
[16]
Python


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measurement_values = [record["measurement_value"] for record in student_records]
coord = (seed_digit, id_checksum, vector_dim)
print("System Config:", sys_config)
print("Generated Records:")
for record in student_records:
print(record)
print("Top Score:", top_score)
print("Sorted Scores:", sorted_scores)
print("Average Score:", round(average_score, 2))
print("Unique Sections:", unique_sections)
print("Measurement Values:", measurement_values)
print("Coordinates:", coord)
[16]
Python
System Config: {'operator': 'Damoluan', 'auth_id': 'TUPM-26-3518', 'base_seed': 8, 'checksum': 25, 'vector_dimension': 8,
Generated Records:
{'operator': 'Damoluan', 'student_id': 'TUPM-26-3518', 'name': 'Alice', 'score': 101, 'section': 'A', 'measurement_catego
{'operator': 'Damoluan', 'student_id': 'TUPM-26-3518', 'name': 'Bob', 'score': 108, 'section': 'B', 'measurement_category
{'operator': 'Damoluan', 'student_id': 'TUPM-26-3518', 'name': 'Charlie', 'score': 115, 'section': 'C', 'measurement_cate
{'operator': 'Damoluan', 'student_id': 'TUPM-26-3518', 'name': 'Diana', 'score': 62, 'section': 'D', 'measurement_categor
{'operator': 'Damoluan', 'student_id': 'TUPM-26-3518', 'name': 'Evan', 'score': 69, 'section': 'A', 'measurement_category
{'operator': 'Damoluan', 'student_id': 'TUPM-26-3518', 'name': 'Fiona', 'score': 76, 'section': 'B', 'measurement_categor
{'operator': 'Damoluan', 'student_id': 'TUPM-26-3518', 'name': 'George', 'score': 83, 'section': 'C', 'measurement_catego
{'operator': 'Damoluan', 'student_id': 'TUPM-26-3518', 'name': 'Hannah', 'score': 90, 'section': 'D', 'measurement_catego


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Select Kernel
print("Top Score:", top_score)
print("Sorted Scores:", sorted_scores)
print("Average Score:", round(average_score, 2))
print("Unique Sections:", unique_sections)
print("Measurement Values:", measurement_values)
print("Coordinates:", coord)
[16]
Python
System Config: {'operator': 'Damoluan', 'auth_id': 'TUPM-26-3518', 'base_seed': 8, 'checksum': 25, 'vector_dimension': 8,
Generated Records:
{'operator': 'Damoluan', 'student_id': 'TUPM-26-3518', 'name': 'Alice', 'score': 101, 'section': 'A', 'measurement_catego
{'operator': 'Damoluan', 'student_id': 'TUPM-26-3518', 'name': 'Bob', 'score': 108, 'section': 'B', 'measurement_category
{'operator': 'Damoluan', 'student_id': 'TUPM-26-3518', 'name': 'Charlie', 'score': 115, 'section': 'C', 'measurement_cate
{'operator': 'Damoluan', 'student_id': 'TUPM-26-3518', 'name': 'Diana', 'score': 62, 'section': 'D', 'measurement_categor
{'operator': 'Damoluan', 'student_id': 'TUPM-26-3518', 'name': 'Evan', 'score': 69, 'section': 'A', 'measurement_category
{'operator': 'Damoluan', 'student_id': 'TUPM-26-3518', 'name': 'Fiona', 'score': 76, 'section': 'B', 'measurement_categor
{'operator': 'Damoluan', 'student_id': 'TUPM-26-3518', 'name': 'George', 'score': 83, 'section': 'C', 'measurement_catego
{'operator': 'Damoluan', 'student_id': 'TUPM-26-3518', 'name': 'Hannah', 'score': 90, 'section': 'D', 'measurement_catego
Top Score: 115
Sorted Scores: [115, 108, 101, 90, 83, 76, 69, 62]
Average Score: 88.0
Unique Sections: ['A', 'B', 'C', 'D']
Measurement Values: [192, 207, 222, 177, 192, 207, 222, 237]
Coordinates: (8, 25, 8)
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organized_records = {}
for record in student_records:
section = record["section"]
if section not in organized_records:
organized_records[section] = []
organized_records[section].append({
"name": record["name"],
"score": record["score"],
"measurement_category": record["measurement_category"],
"measurement_value": record["measurement_value"],
"sensor": record["sensor"]
})
print("Organized Records by Section:")
for section, entries in organized_records.items():
print(f"\nSection {section}:")
for entry in entries:
print(entry)
[19]
Python
Organized Records by Section:
Section A:
{'name': 'Alice', 'score': 101, 'measurement_category': 'TEMPERATURE', 'measurement_value': 192, 'sensor': 'Sensor-1'}
{'name': 'Evan', 'score': 69, 'measurement_category': 'SPEED', 'measurement_value': 192, 'sensor': 'Sensor-5'}


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"measurement_value": record["measurement_value"],
"sensor": record["sensor"]
})
print("Organized Records by Section:")
for section, entries in organized_records.items():
print(f"\nSection {section}:")
for entry in entries:
print(entry)
[19]
Python
Organized Records by Section:
Section A:
{'name': 'Alice', 'score': 101, 'measurement_category': 'TEMPERATURE', 'measurement_value': 192, 'sensor': 'Sensor-1'}
{'name': 'Evan', 'score': 69, 'measurement_category': 'SPEED', 'measurement_value': 192, 'sensor': 'Sensor-5'}
Section B:
{'name': 'Bob', 'score': 108, 'measurement_category': 'VOLTAGE', 'measurement_value': 207, 'sensor': 'Sensor-2'}
{'name': 'Fiona', 'score': 76, 'measurement_category': 'TEMPERATURE', 'measurement_value': 207, 'sensor': 'Sensor-6'}
Section C:
{'name': 'Charlie', 'score': 115, 'measurement_category': 'PRESSURE', 'measurement_value': 222, 'sensor': 'Sensor-3'}
{'name': 'George', 'score': 83, 'measurement_category': 'VOLTAGE', 'measurement_value': 222, 'sensor': 'Sensor-7'}
Section D:
{'name': 'Diana', 'score': 62, 'measurement_category': 'FLOW', 'measurement_value': 177, 'sensor': 'Sensor-4'}
{'name': 'Hannah', 'score': 90, 'measurement_category': 'PRESSURE', 'measurement_value': 237, 'sensor': 'Sensor-8'}
Generate
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```python
# Appendix C
section_summary = {}

for section, entries in organized_records.items():
    scores = [entry["score"] for entry in entries]
    measurements = [entry["measurement_value"] for entry in entries]
    categories = sorted({entry["measurement_category"] for entry in entries})

    section_summary[section] = {
        "students": len(entries),
        "average_score": round(sum(scores) / len(scores), 2),
        "top_score": max(scores),
        "average_measurement": round(sum(measurements) / len(measurements), 2),
        "categories": categories
    }

print("Performance Summary by Section:")
for section, summary in section_summary.items():
    print(f"{section}: {summary}")

top_performer = max(student_records, key=lambda record: record["score"])
print("\nTop Performer:")
print(top_performer)

highest_section = max(section_summary.items(), key=lambda item: item[1]["average_score"])
print("\nHighest Average Section Performance:")
print(highest_section)
```

[18] Python

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```
    "average_measurement": round(sum(measurements) / len(measurements), 2),
    "categories": categories
}

print("Performance Summary by Section:")
for section, summary in section_summary.items():
    print(f"{section}: {summary}")

top_performer = max(student_records, key=lambda record: record["score"])
print("\nTop Performer:")
print(top_performer)

highest_section = max(section_summary.items(), key=lambda item: item[1]["average_score"])
print("\nHighest Average Section Performance:")
print(highest_section)
```

[18] Python

```
Performance Summary by Section:
A: {'students': 2, 'average_score': 85.0, 'top_score': 101, 'average_measurement': 192.0, 'categories': ['SPEED', 'TEMPER
B: {'students': 2, 'average_score': 92.0, 'top_score': 108, 'average_measurement': 207.0, 'categories': ['TEMPERATURE', '
C: {'students': 2, 'average_score': 99.0, 'top_score': 115, 'average_measurement': 222.0, 'categories': ['PRESSURE', 'VOL
D: {'students': 2, 'average_score': 76.0, 'top_score': 90, 'average_measurement': 207.0, 'categories': ['FLOW', 'PRESSURE

Top Performer:
{'operator': 'Damoluan', 'student_id': 'TUPM-26-3518', 'name': 'Charlie', 'score': 115, 'section': 'C', 'measurement_cate

Highest Average Section Performance:
('C', {'students': 2, 'average_score': 99.0, 'top_score': 115, 'average_measurement': 222.0, 'categories': ['PRESSURE', '
```